

Sumaya Tabassum
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SUMMARY

- Environmental and Civil Engineering professional focused on wastewater contaminant removal and sustainable construction, seeking opportunities in municipal water and wastewater design.
- Developed an innovative and cost-effective PFAS remediation strategy by modifying coal fly ash (CFA) to adsorb PFAS spiked into real wastewater and landfill leachate, addressing practical treatment challenges.
- Incorporated PFAS-laden CFA into concrete as a partial cement replacement, promoting circular economy principles while reducing long-term environmental contamination risks.
- Evaluated PFAS retention and leaching behavior through EPA 1315 leaching tests and compressive strength analysis, ensuring structural stability and safe disposal of PFAS-embedded concrete.
- Investigated irrigation and drinking water quality in Bangladesh by analyzing physicochemical parameters and bacterial contamination, benchmarking against WHO and national standards.
- Committed to integrating waste reuse into sustainable engineering practices, contributing to resource recovery and the development of environmentally resilient infrastructure.
- EIT candidate with strong research, problem-solving, and communication skills, capable of translating complex environmental data into practical engineering solutions.

EDUCATION

- **M.Sc. in Civil Engineering (Major Environmental Engineering) (GPA: 4.0)** **Fall - 2025 (Expected)**
North Carolina A&T State University (NCAT), NC, USA
- **M.Sc. in Soil, Water and Environment (GPA: 3.55)** **Dec 2022**
Khulna University, Bangladesh
- **B.Sc. in Soil, Water and Environment (GPA: 3.23)** **Jan 2020**
Khulna University, Bangladesh

SKILLS & TRAINING

Combustion Ion Chromatography (CIC)	Microsoft Excel	AutoCAD
TOC analyzer	Microsoft Word	Origin Lab
Spectrophotometry	Microsoft PowerPoint	ANOVA
EPA 1315 Leaching Test		

WORK EXPERIENCES

- Graduate Research and Teaching Assistant, CAEE, NCAT** **January 2024 – Present**
- Modified coal fly ash (CFA) through acid and thermal treatments to enhance its PFAS adsorption capacity from real wastewater, pure solutions, and landfill leachate, targeting practical applications in environmental remediation.
 - Designed and optimized concrete mixes incorporating PFAS-laden CFA as a partial cement substitute, and conducted both dry and wet curing processes to ensure structural performance.
 - Performed compressive strength testing and EPA 1315 leaching assessments on concrete specimens to evaluate long-term PFAS retention and environmental safety.
 - Analyzed PFAS leaching behavior using Combustion Ion Chromatography (CIC) and total organic carbon quantification via TOC analyzer, calculating leachability indices and understanding PFAS transport mechanisms.
 - Mentored two undergraduate researchers in laboratory techniques, data analysis, and scientific communication, fostering collaborative and skill-based learning. Collaborated with lab teams and reported data to supervisors—skills aligned with multidisciplinary design coordination.
 - Presented research outcomes at the AEESP Conference and university research symposium, and provided consistent project updates to supervisors to support informed decision-making and timeline adherence.

- **Teaching Assistant (TA)** – Assisted students with lab experiments and problem-solving, ensuring understanding of environmental concepts. Managed grading, provided feedback, and facilitated discussions to enhance learning.
- Led sustainable research initiatives, integrating waste repurposing into concrete design to reduce PFAS contamination risks.

Assistant Teacher, Govt. Primary School of Bangladesh

Jan 2023 – May 2023

- Developed lesson plans for reading, science, and math.
- Taught and assessed students' social interaction skills, identifying strengths and areas for improvement. Instructed classes and small groups, grading assignments to evaluate comprehension.

Assistant Officer – IFIC Bank PLC; Dhaka, Bangladesh

Oct 2021 – Jan 2023

- Creating, managing, and maintaining accounts. Handling customer interactions, providing product information, and opening accounts.
- Offering a one-stop banking service and managing loans. Inputting and collating data from various sources, including purchasing card facilities and asset registers

SELECTED PUBLICATIONS

- **Tabassum, S.***, Dristy, J. T., Ahmed, A., & Riyad, R. H. (2024). Assessment of tube well water quality in selected residential areas in Khulna. *Environmental Research and Technology*, 7(1), 41–60. <https://doi.org/10.35208/ert.1362577>.
- **Tabassum, S.***. (2024). An Investigation into the Efficacy of Septic Tank Systems in Removing Organics in a Subtropical Climate. *SVU-International Journal of Engineering Sciences and Applications*, 5(1), 125–135. <https://doi.org/10.21608/svusrc.2023.247561.1160>
- **Tabassum, S.**, Tasnim, S., Islam, N., Islam, T. N., & Dev, A. (n.d.). Environmental consequences of dumping dead batteries of" instant power supply unit (IPS) on surface water: a case study on Khulna city. September 11, 2024, http://www.iccesd.com/proc_2024/Papers/257.pdf

AWARDS, TRAINING, WORKSHOPS, AND FIELD TRIPS

- **Best University-Wide Outstanding Master's Graduate Research Assistant Award** 2025, NCAT
- **Best Poster Award** at COE Graduate Poster Competition 2025, NCAT
- Global Environmental Management - Technical University of Denmark (Coursera NFNU46J9UQXK)
- Understanding Research Methods - University of London (Coursera NFUUNWJXDY4V)
- Career Development Workshop of Nitol-Niloy Group.
- General Lab Safety and Hazardous Waste Training
- Field visits to the T. Z. Osborne Wastewater Treatment Plant and Mitchell Drinking Water Plant, gaining practical exposure to WWTP and drinking water operations.

LEADERSHIP EXPERIENCE

- Joint Secretary at Science Club of Chuadanga; organizing Physics and Math Olympiad 2018 - 2020
- Executive Member of the Soil, Water and Environment discipline association, KU 2017-2020
Responsible for organizing fresher's reception, cultural, and farewell events
- Member of Pedon; Soil, Water and Environment Discipline 2018 -2022
- Member of Badhon, blood donation group of Khulna University, 2018 -2022
- Former Sports Secretary, Soil, Water and Environment Discipline, KU 2018 -2020

HOBBIES AND INTERESTS

- Hobbies: Travelling, Gardening, Photography
- Volunteering (Worked for underserved communities and donated blood)